ModelIt!

Student Activity Pack

Why Can't You See Your Own Cells?

Exploring the Microscopic Building Blocks of Life

Grade 6 | NGSS MS-LS1-1, MS-LS1-2 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think cells are? Where are they found?

2 Why do you think you need a microscope to see cells?

3 Draw what you think a cell looks like inside.

4 If cells are alive, what do they need to survive?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Cell Size | Nutrient Supply | Oxygen Level | Waste Buildup

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy Cell

Moderate nutrient supply, normal oxygen

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Starvation Mode

Lock Nutrient Supply to 10%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Oxygen Crisis

Set Oxygen Level to 15%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cell Division Rate — How quickly cells copy themselves to grow or repair tissue

Temperature — Temperature affects how fast chemical reactions happen inside cells

pH Level — The acidity of the cell's environment affects enzyme function

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If your body is made of 37 trillion cells, why can't you see a single one? Use evidence from your simulation to support your answers.

1 Why can't a single cell grow to be the size of a basketball?

2 How is a cell like a tiny factory? What are the 'machines' inside?

3 What would happen to your body if your cells couldn't get rid of waste?

4 Why do different cells in your body have different shapes and sizes?

5 How does your model help explain something you can't see with your eyes?

8 STEM Challenge: Design a Cell City

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YOUR ENGINEERING MISSION

Design a model city where each building represents a cell organelle, showing how resources flow in and waste flows out — just like a real cell.

THE SCENARIO

Your city planning team has been hired to design 'Cell City' — a community where every structure has a job, resources must reach every building, and waste removal is critical. Use your model to decide the best layout.

Engineering Design Thinking

1. WHAT STRUCTURES WILL REPRESENT THE CELL MEMBRANE, NUCLEUS, AND MITOCHONDRIA?
2. HOW WILL RESOURCES (NUTRIENTS, OXYGEN) BE DELIVERED THROUGHOUT YOUR CITY?
3. WHAT WASTE REMOVAL SYSTEMS WILL YOU INCLUDE?
4. HOW WILL YOU SHOW THAT EVERY PART OF THE CITY DEPENDS ON EVERY OTHER PART?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Cell Biologists and Biomedical Researchers use powerful microscopes and computer models to study cells and develop treatments for diseases. They earn $60,000–$120,000/year.